

Week 2 Lecture Note

Topic: Understand the idea of CNNs and build a small Sequential model in Google Colab

1. Objectives

- (1) Explain what a CNN does in simple language: it learns useful patterns from images.
- (2) Introduce the role of the main layers: Conv2D, MaxPooling, Flatten, and Dense.
- (3) Help students assemble a basic Sequential model without writing everything from scratch.

2. Key Concepts

(1) What is a CNN?

A Convolutional Neural Network (CNN) is a deep learning model designed for image analysis. It detects small visual patterns first, then combines them to make a final prediction such as "Pneumonia" or "Normal."

In simple terms, the process is: find features → reduce size → flatten information → make a prediction

(2) Conv2D with Filter / Kernel

A filter (or kernel) is like a small moving window. It scans the image step by step and responds when it finds useful patterns such as edges, brightness changes, or textures.

Instead of understanding the whole image at once, the model examines small regions and extracts important features gradually.

(3) Pooling

Pooling reduces the size of the feature maps while keeping the most important information. This makes the model more efficient and helps it focus on stronger signals. **MaxPooling** keeps the strongest responses and removes less important details, reducing computation.

(4) Dense Layer

After the model extracts image features, the Dense layer combines them and makes the final decision.

Before this step, the 2D feature maps are converted into a 1D form so the model can use all collected clues to classify the image as Normal or Pneumonia.

(5) Compile: The compile step tells the model how to learn.

Loss: measures how wrong the prediction is

Optimizer: updates the model to reduce error

Metrics: shows performance during training